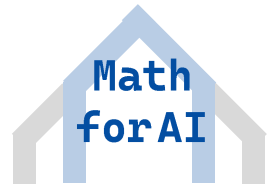




충남대학교



2026년 2학기

AI를 위한 수학

Linear Algebra

임성수

목차

1. Scalars, Vectors, Matrices and Tensors
2. Multiplying Matrices and Vectors
3. Identify and Inverse Matrices
4. Linear Dependence and Span
5. Norms
6. Special Kinds of Matrices and Vectors
7. Eigendecomposition
8. Singular Value Decomposition
9. The Moore–Penrose Pseudoinverse
10. The Trace Operator
11. The Determinant
12. Principal Components Analysis



1. Scalars, Vectors, Matrices and Tensors

선형대수학에서 다루는 수학적 대상들

- 스칼라(scalar): 하나의 수
- 벡터(vector): 수들의 배열(array)
- 행렬(matrix): 수들의 2차원 배열
- 텐서(tensor): 수들의 다차원 배열

(11)

SCALAR

5 3 7

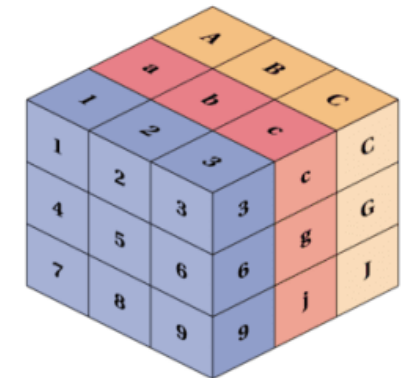
Row Vector
(shape 1x3)

5
1.5
2

Column Vector
(shape 3x1)

4 19 8
16 3 5

MATRIX



TENSOR

1. Scalars, Vectors, Matrices and Tensors

설명

- 스칼라(scalar): 실수($\mathbb{R}$) 혹은 자연수($\mathbb{N}$)와 같은 수를 표현, $s \in \mathbb{R}$, $n \in \mathbb{N}$

- 벡터(vector): n 차원 벡터 $\mathbf{x} = (x_1, \dots, x_n) = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$

$\mathbf{x}$ 의 인덱스(index) i 마다 원소(element) x_i 포함

집합(set) $S = \{i_1, \dots, i_{|S|}\}$ 에 대해, $\mathbf{x}_S = (x_{i_1}, \dots, x_{i_{|S|}}) \in \mathbb{R}^{|S|}$

- 벡터의 연산: 벡터합(vector addition): $\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n : \mathbf{x} + \mathbf{y} = (x_1 + y_1, \dots, x_n + y_n)$
스칼라곱(scalar multiplication): $\forall \lambda \in \mathbb{R}, \mathbf{x} \in \mathbb{R}^n : \lambda \mathbf{x} = \lambda(x_1, \dots, x_n) = (\lambda x_1, \dots, \lambda x_n)$

1. Scalars, Vectors, Matrices and Tensors

설명

- 행렬(matrix): $m \times n$ 행렬, $A = \begin{bmatrix} A_{1,1} & \cdots & A_{1,n} \\ \vdots & \ddots & \vdots \\ A_{m,1} & \cdots & A_{m,n} \end{bmatrix} \in \mathbb{R}^{m \times n}$

A 의 튜플(tuple) (i, j) 마다 원소 $A_{i,j}$ 포함

A 의 i 번째 행(row) $A_{i,:} \in \mathbb{R}^n$, A 의 j 번째 열(column) $A_{:,j} \in \mathbb{R}^m$

행렬값(matrix-valued) 함수 f 의 (i, j) 번째 항(entry): $f(A)_{i,j}$

- 텐서(tensor): A 의 좌표(coordinate) $(i_1, \dots, i_k)$ 마다 원소 $A_{i_1, \dots, i_k}$ 포함

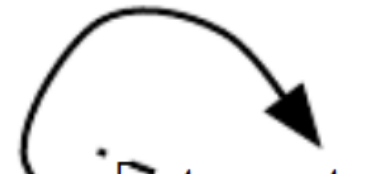
1. Scalars, Vectors, Matrices and Tensors

행렬의 기초 연산

- 전치(transpose): 행렬의 주요 연산, 행과 열을 바꾸는 연산
 행렬 A 에 대해 A^T 는 다음과 같이 정의: $(A^T)_{i,j} = A_{j,i}$

예제

•



$$A = \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \\ A_{3,1} & A_{3,2} \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} A_{1,1} & A_{2,1} & A_{3,1} \\ A_{1,2} & A_{2,2} & A_{3,2} \end{bmatrix}$$

1. Scalars, Vectors, Matrices and Tensors

행렬의 기초 연산

- 행렬합(matrix addition): $\mathbf{A}$, $\mathbf{B}$ 의 행, 열의 개수가 같을 때, $\mathbf{C} = \mathbf{A} + \mathbf{B}$ where $C_{i,j} = A_{i,j} + B_{i,j}$
- 스칼라곱(scalar multiplication): $\mathbf{D} = a \cdot \mathbf{B}$ where $D_{i,j} = a \cdot B_{i,j}$

예제

- $\mathbf{D} = a \cdot \mathbf{B} + c$ where $D_{i,j} = a \cdot B_{i,j} + c$
- 브로드캐스팅(broadcasting): $\mathbf{C} = \mathbf{A} + \mathbf{b}$ where $C_{i,j} = A_{i,j} + B_j$

2. Multiplying Matrices and Vectors

벡터의 곱셈

- 점곱/스칼라곱(dot product, scalar product):
같은 차원(dimension)의 벡터들 $\mathbf{x}, \mathbf{y}$ 에 대해 $\mathbf{x}^\top \mathbf{y} = \sum_i x_i y_i$

행렬의 곱셈

- 행렬곱(matrix product):
 $\mathbf{A}$ 의 행의 개수와 $\mathbf{B}$ 의 열의 개수가 같을 때, $\mathbf{C} = \mathbf{AB}$ where $C_{i,j} = \sum_k A_{i,k} B_{k,j}$
- 아다마르곱/원소별곱(Hadamard product, element-wise product):
 $\mathbf{A}, \mathbf{B}$ 의 행, 열의 개수가 같을 때, $\mathbf{C} = \mathbf{A} \odot \mathbf{B}$ where $C_{i,j} = A_{i,j} \cdot B_{i,j}$

2. Multiplying Matrices and Vectors

곱셈의 법칙

- 결합법칙(associativity):
 $\forall \mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{B} \in \mathbb{R}^{n \times p} : (\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$
- 분배법칙(distributivity):
 $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}, \mathbf{C}, \mathbf{D} \in \mathbb{R}^{n \times p} : (\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}, \mathbf{A}(\mathbf{C} + \mathbf{D}) = \mathbf{AC} + \mathbf{AD}$
- 교환법칙(commutative):
 행렬곱에서 성립 안 함 $\mathbf{AB} \neq \mathbf{BA}$
- 전치 관련 법칙들:
 $\mathbf{x}^\top \mathbf{y} = \mathbf{y}^\top \mathbf{x}$
 $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$

2. Multiplying Matrices and Vectors

예제

Example 2.3

For $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 3}$, $\mathbf{B} = \begin{bmatrix} 0 & 2 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 2}$, we obtain

$$\mathbf{AB} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 2 & 5 \end{bmatrix} \in \mathbb{R}^{2 \times 2}, \quad (2.15)$$

$$\mathbf{BA} = \begin{bmatrix} 0 & 2 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 4 & 2 \\ -2 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 3}. \quad (2.16)$$

2. Multiplying Matrices and Vectors

선형시스템(linear system, system of linear equations): 선형연립방정식

- $$\begin{cases} A_{1,1}x_1 + \cdots + A_{1,n}x_n = b_1 \\ \vdots \\ A_{m,1}x_1 + \cdots + A_{m,n}x_n = b_m \end{cases}$$
 을 만족하는 해(solution) $(x_1, \dots, x_n)$ 을 구하는 문제
- $A\mathbf{x} = \mathbf{b}$ where $A \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{x} \in \mathbb{R}^n$

선형시스템의 풀이

- $A\mathbf{x} = \mathbf{b}$ 의 특수해(particular solution) 구하기
- $A\mathbf{x} = \mathbf{0}$ 의 모든 해 또는 영공간(null space) 구하기
- 위 두 과정의 해를 합쳐서 일반해(general solution) 구하기

Example 2.6

For $a \in \mathbb{R}$, we seek all solutions of the following system of equations:

$$\begin{array}{rrrrrrrcl} -2x_1 & + & 4x_2 & - & 2x_3 & - & x_4 & + & 4x_5 & = & -3 \\ 4x_1 & - & 8x_2 & + & 3x_3 & - & 3x_4 & + & x_5 & = & 2 \\ x_1 & - & 2x_2 & + & x_3 & - & x_4 & + & x_5 & = & 0 \\ x_1 & - & 2x_2 & & & & -3x_4 & + & 4x_5 & = & a \end{array} \quad (2.44)$$

We start by converting this system of equations into the compact matrix notation $\mathbf{A}\mathbf{x} = \mathbf{b}$. We no longer mention the variables $\mathbf{x}$ explicitly and build the *augmented matrix* (in the form $[\mathbf{A} \mid \mathbf{b}]$)

$$\left[\begin{array}{ccccc|c} -2 & 4 & -2 & -1 & 4 & -3 \\ 4 & -8 & 3 & -3 & 1 & 2 \\ 1 & -2 & 1 & -1 & 1 & 0 \\ 1 & -2 & 0 & -3 & 4 & a \end{array} \right] \begin{array}{l} \text{Swap with } R_3 \\ \\ \text{Swap with } R_1 \end{array}$$

where we used the vertical line to separate the left-hand side from the right-hand side in (2.44). We use $\rightsquigarrow$ to indicate a transformation of the augmented matrix using elementary transformations.

예제

예제

Swapping Rows 1 and 3 leads to

$$\left[\begin{array}{ccccc|c} 1 & -2 & 1 & -1 & 1 & 0 \\ 4 & -8 & 3 & -3 & 1 & 2 \\ -2 & 4 & -2 & -1 & 4 & -3 \\ 1 & -2 & 0 & -3 & 4 & a \end{array} \right] \begin{array}{l} \\ -4R_1 \\ +2R_1 \\ -R_1 \end{array}$$

When we now apply the indicated transformations (e.g., subtract Row 1 four times from Row 2), we obtain

$$\begin{array}{l} \left[\begin{array}{ccccc|c} 1 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 & -3 & 2 \\ 0 & 0 & 0 & -3 & 6 & -3 \\ 0 & 0 & -1 & -2 & 3 & a \end{array} \right] \begin{array}{l} \\ \\ -R_2 - R_3 \end{array} \\ \\ \rightsquigarrow \left[\begin{array}{ccccc|c} 1 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 & -3 & 2 \\ 0 & 0 & 0 & -3 & 6 & -3 \\ 0 & 0 & 0 & 0 & 0 & a+1 \end{array} \right] \begin{array}{l} \\ \cdot(-1) \\ \cdot(-\frac{1}{3}) \\ \end{array} \\ \\ \rightsquigarrow \left[\begin{array}{ccccc|c} 1 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 3 & -2 \\ 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 & a+1 \end{array} \right] \end{array}$$

This (augmented) matrix is in a convenient form, the *row-echelon form* (REF). Reverting this compact notation back into the explicit notation with the variables we seek, we obtain

$$\begin{array}{ccccccccc} x_1 & - & 2x_2 & + & x_3 & - & x_4 & + & x_5 & = & 0 \\ & & & & x_3 & - & x_4 & + & 3x_5 & = & -2 \\ & & & & & & x_4 & - & 2x_5 & = & 1 \\ & & & & & & & & 0 & = & a + 1 \end{array} \quad . \quad (2.45)$$

Only for $a = -1$ this system can be solved. A *particular solution* is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix} . \quad (2.46)$$

The *general solution*, which captures the set of all possible solutions, is

$$\left\{ \boldsymbol{x} \in \mathbb{R}^5 : \boldsymbol{x} = \begin{bmatrix} 2 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 2 \\ 0 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \quad \lambda_1, \lambda_2 \in \mathbb{R} \right\} . \quad (2.47)$$

예제

3. Identify and Inverse Matrices

행렬의 역

- 항등행렬(identity matrix): n 차원 항등행렬 I_n (또는 I)은 다음을 만족, $\forall x \in \mathbb{R}^n : I_n x = x$
- 역행렬(inverse matrix): $A^{-1}A = AA^{-1} = I_n$ 을 만족하는 행렬 A^{-1}
- 가역행렬(invertible matrix): 역행렬이 존재하는 정방(square) 행렬 A
- A 가 가역일 때, 선형시스템 $Ax = b \Leftrightarrow A^{-1}Ax = A^{-1}b \Leftrightarrow x = A^{-1}b$
- A 가 정방행렬이 아닐 때, $Ax = b \Leftrightarrow A^T Ax = A^T b \Leftrightarrow x = (A^T A)^{-1} A^T b$

예제

- $$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3. Identify and Inverse Matrices

예제

Example 2.9 (Calculating an Inverse Matrix by Gaussian Elimination)

To determine the inverse of

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad (2.57)$$

we write down the augmented matrix

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{array} \right]$$

3. Identify and Inverse Matrices

예제

and use Gaussian elimination to bring it into reduced row-echelon form

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -1 & 2 & -2 & 2 \\ 0 & 1 & 0 & 0 & 1 & -1 & 2 & -2 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 & -1 & 0 & -1 & 2 \end{array} \right],$$

such that the desired inverse is given as its right-hand side:

$$\mathbf{A}^{-1} = \begin{bmatrix} -1 & 2 & -2 & 2 \\ 1 & -1 & 2 & -2 \\ 1 & -1 & 1 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}. \quad (2.58)$$

We can verify that (2.58) is indeed the inverse by performing the multiplication $\mathbf{A}\mathbf{A}^{-1}$ and observing that we recover $\mathbf{I}_4$.

4. Linear Dependence and Span

선형결합

- 선형결합(linear combination): 벡터들의 집합 $\{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(n)}\}$, 상수 $c_1, \dots, c_n$ 에 대해 $\sum_i c_i \mathbf{v}^{(i)}$
- 생성(span): 벡터 집합 $\{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(n)}\}$ 에 대해 모든 가능한 선형결합의 집합 $\{\sum_i c_i \mathbf{v}^{(i)} \mid c_1, \dots, c_n \in \mathbb{R}\}$

선형결합과 선형시스템

- 열공간(column space) 또는 치역(range): $\{\mathbf{Ax} \mid \mathbf{x} \in \mathbb{R}^n\} = \{\sum_i x_i \mathbf{A}_{:,i} \mid x_1, \dots, x_n \in \mathbb{R}\} \subseteq \mathbb{R}^m$
- 일치(consistent): 선형시스템 $\mathbf{Ax} = \mathbf{b}$ 의 해가 존재함은 $\mathbf{A}$ 의 열공간이 $\mathbf{b}$ 를 포함하는 것을 의미
- 선형시스템의 해는 없거나, 유일하거나, 무한히 많을 수 있음

4. Linear Dependence and Span

예제

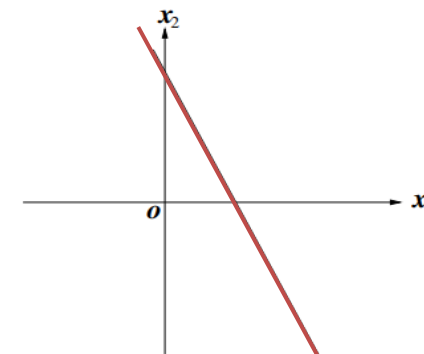
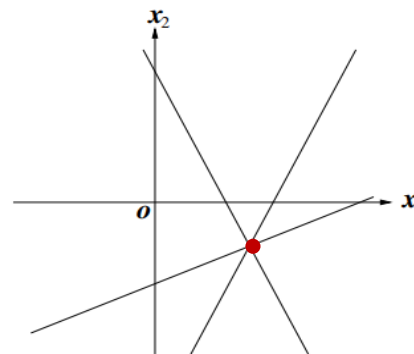
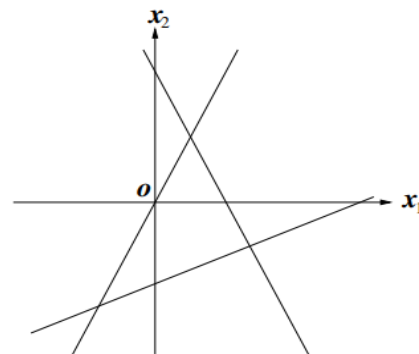
The following three linear systems

$$(a) \begin{cases} 2x_1 + x_2 = 3 \\ 2x_1 - x_2 = 0 \\ x_1 - 2x_2 = 4 \end{cases}$$

$$(b) \begin{cases} 2x_1 + x_2 = 3 \\ 2x_1 - x_2 = 5 \\ x_1 - 2x_2 = 4 \end{cases}$$

$$(c) \begin{cases} 2x_1 + x_2 = 3 \\ 4x_1 + 2x_2 = 6 \\ 6x_1 + 3x_2 = 9 \end{cases}$$

has no solution, a unique solution, and infinitely many solutions, resp.



4. Linear Dependence and Span

선형독립

- 선형독립(linearly independent): 어떠한 벡터도 다른 벡터들의 선형결합으로 표현 불가능한 경우
- m 차원 벡터 n 개가 선형독립이라면 각 벡터를 열에 배치한 행렬의 열공간의 차원은 n
- 선형종속(linearly dependent): 선형독립의 반대

선형독립과 역행렬

- m 차원 벡터 m 개가 선형독립이라면 각 벡터를 열에 배치한 행렬의 열공간은 $\mathbb{R}^m$
- 이러한 행렬은 정방(square) 행렬이고, 역행렬이 존재함(nonsingular, invertible)
- $m \times m$ 정방 행렬의 열들이 선형종속이라면, 역행렬이 존재하지 않음(singular)

4. Linear Dependence and Span

예제

Example 2.14

Consider $\mathbb{R}^4$ with

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ -3 \\ 4 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} -1 \\ -2 \\ 1 \\ 1 \end{bmatrix}. \quad (2.67)$$

To check whether they are linearly dependent, we follow the general approach and solve

$$\lambda_1 \mathbf{x}_1 + \lambda_2 \mathbf{x}_2 + \lambda_3 \mathbf{x}_3 = \lambda_1 \begin{bmatrix} 1 \\ 2 \\ -3 \\ 4 \end{bmatrix} + \lambda_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix} + \lambda_3 \begin{bmatrix} -1 \\ -2 \\ 1 \\ 1 \end{bmatrix} = \mathbf{0} \quad (2.68)$$

4. Linear Dependence and Span

예제

for $\lambda_1, \dots, \lambda_3$. We write the vectors $\mathbf{x}_i$, $i = 1, 2, 3$, as the columns of a matrix and apply elementary row operations until we identify the pivot columns:

$$\begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & -2 \\ -3 & 0 & 1 \\ 4 & 2 & 1 \end{bmatrix} \rightsquigarrow \dots \rightsquigarrow \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}. \quad (2.69)$$

Here, every column of the matrix is a pivot column. Therefore, there is no non-trivial solution, and we require $\lambda_1 = 0$, $\lambda_2 = 0$, $\lambda_3 = 0$ to solve the equation system. Hence, the vectors $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are linearly independent.

5. Norms

벡터의 크기

- 크기 또는 노름(norm): 벡터공간(vector space) V 에 대한 함수 $\|\cdot\|: V \rightarrow \mathbb{R}$, 아래 성질 만족
 절대동차(absolutely homogeneous): $\|\lambda \mathbf{x}\| = |\lambda| \|\mathbf{x}\|$
 삼각부등식(triangle inequality): $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$
 양의 정부호(positive definite): (i) $\|\mathbf{x}\| \geq 0$, (ii) $\|\mathbf{x}\| = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}$
- L^p 노름: $\|\mathbf{x}\|_p = (\sum_i |x_i|^p)^{\frac{1}{p}}$
- 최대(max) 노름: $\|\mathbf{x}\|_\infty = \max_i |x_i|$

행렬의 크기

- Frobenius 노름: $\|\mathbf{A}\|_F = \sqrt{\sum_{i,j} A_{i,j}^2}$ (벡터의 L^2 노름의 확장)

예제

Example 3.1 (Manhattan Norm)

The *Manhattan norm* on $\mathbb{R}^n$ is defined for $\mathbf{x} \in \mathbb{R}^n$ as

$$\|\mathbf{x}\|_1 := \sum_{i=1}^n |x_i|, \quad (3.3)$$

where $|\cdot|$ is the absolute value. The left panel of Figure 3.3 shows all vectors $\mathbf{x} \in \mathbb{R}^2$ with $\|\mathbf{x}\|_1 = 1$. The Manhattan norm is also called ℓ_1 norm.

Example 3.2 (Euclidean Norm)

The *Euclidean norm* of $\mathbf{x} \in \mathbb{R}^n$ is defined as

$$\|\mathbf{x}\|_2 := \sqrt{\sum_{i=1}^n x_i^2} = \sqrt{\mathbf{x}^\top \mathbf{x}} \quad (3.4)$$

and computes the *Euclidean distance* of $\mathbf{x}$ from the origin. The right panel of Figure 3.3 shows all vectors $\mathbf{x} \in \mathbb{R}^2$ with $\|\mathbf{x}\|_2 = 1$. The Euclidean norm is also called ℓ_2 norm.

5. Norms

코사인 유사도

- 두 벡터 $\mathbf{x}, \mathbf{y}$ 사이의 각 w 에 대해, $\cos w = \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}$

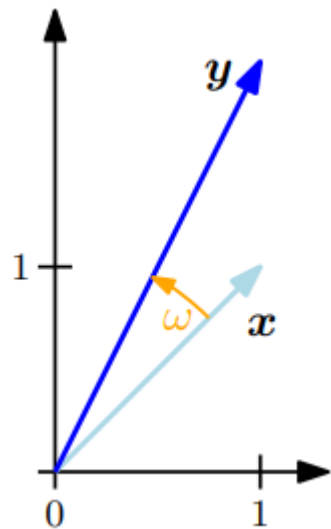
예제

Example 3.6 (Angle between Vectors)

Let us compute the angle between $\mathbf{x} = [1, 1]^\top \in \mathbb{R}^2$ and $\mathbf{y} = [1, 2]^\top \in \mathbb{R}^2$; see Figure 3.5, where we use the dot product as the inner product. Then we get

$$\cos \omega = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\sqrt{\langle \mathbf{x}, \mathbf{x} \rangle \langle \mathbf{y}, \mathbf{y} \rangle}} = \frac{\mathbf{x}^\top \mathbf{y}}{\sqrt{\mathbf{x}^\top \mathbf{x} \mathbf{y}^\top \mathbf{y}}} = \frac{3}{\sqrt{10}}, \quad (3.26)$$

and the angle between the two vectors is $\arccos(\frac{3}{\sqrt{10}}) \approx 0.32$ rad, which corresponds to about 18° .



5. Norms

벡터의 공간

- 벡터공간(vector space):
벡터 집합 V 와 연산 벡터합($+$: $V \times V \rightarrow V$), 스칼라곱($\cdot$: $\mathbb{R} \times V \rightarrow V$)으로 정의
- 벡터공간이 만족해야 하는 주요 조건:
 분배법칙(distributivity): $\forall \lambda \in \mathbb{R}, \mathbf{x}, \mathbf{y} \in V : \lambda \cdot (\mathbf{x} + \mathbf{y}) = \lambda \cdot \mathbf{x} + \lambda \cdot \mathbf{y}$
 분배법칙(distributivity): $\forall \lambda, \psi \in \mathbb{R}, \mathbf{x} \in V : (\lambda + \psi) \cdot \mathbf{x} = \lambda \cdot \mathbf{x} + \psi \cdot \mathbf{x}$
 결합법칙(associativity): $\forall \lambda, \psi \in \mathbb{R}, \mathbf{x} \in V : \lambda \cdot (\psi \cdot \mathbf{x}) = (\lambda \cdot \psi) \cdot \mathbf{x}$
 항등원(neutral element): $\forall \mathbf{x} \in V : 1 \cdot \mathbf{x} = \mathbf{x}$

5. Norms

벡터의 내적

- 내적(inner product): $\Omega: V \times V \rightarrow \mathbb{R}$ 이 아래 조건들을 만족하면 내적이라 함
 이중선형(bilinearity): $\forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in V, \lambda, \psi \in \mathbb{R} : \Omega(\lambda \mathbf{x} + \psi \mathbf{y}, \mathbf{z}) = \lambda \Omega(\mathbf{x}, \mathbf{z}) + \psi \Omega(\mathbf{y}, \mathbf{z})$
 이중선형(bilinearity): $\forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in V, \lambda, \psi \in \mathbb{R} : \Omega(\mathbf{x}, \lambda \mathbf{y} + \psi \mathbf{z}) = \lambda \Omega(\mathbf{x}, \mathbf{y}) + \psi \Omega(\mathbf{x}, \mathbf{z})$
 대칭(symmetric): $\forall \mathbf{x}, \mathbf{y} \in V : \Omega(\mathbf{x}, \mathbf{y}) = \Omega(\mathbf{y}, \mathbf{x})$
 양의 정부호(positive definite): $\forall \mathbf{x} \in V \setminus \{\mathbf{0}\} : \Omega(\mathbf{x}, \mathbf{x}) > 0, \Omega(\mathbf{0}, \mathbf{0}) = 0$
- $\Omega(\mathbf{x}, \mathbf{y})$ 대신 $\langle \mathbf{x}, \mathbf{y} \rangle$ 로 주로 표기

벡터의 길이

- 길이(length): $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$
- 거리(distance): $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{\langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y} \rangle}$

6. Special Kinds of Matrices and Vectors

대각행렬

- 대각행렬(diagonal matrix): $D_{i,j} = 0$ if $i \neq j$ 를 만족하는 행렬 D
- 항등행렬(identity matrix): $I_{i,j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$ 를 만족하는 행렬 I
- 벡터 x 의 각 항을 대각항(diagonal entry)으로 가지는 대각행렬이 $\text{diag}(v)$ 일 때,
 $\text{diag}(v)x = v \odot x$

$$\text{diag}(v)^{-1} = \text{diag}\left(\left[\frac{1}{v_1}, \dots, \frac{1}{v_n}\right]^T\right)$$

대칭행렬

- 대칭행렬(symmetric matrix): $A = A^T$ 을 만족하는 정방행렬 A , $\forall i, j \in \{1, \dots, n\}: A_{i,j} = A_{j,i}$

6. Special Kinds of Matrices and Vectors

단위벡터

- 단위벡터(unit vector): $\|\mathbf{x}\|_2 = 1$ 인 벡터 $\mathbf{x}$

벡터의 직교성

- 직교(orthogonal, perpendicular): $\mathbf{x}^\top \mathbf{y} = 0$ 인 벡터 $\mathbf{x}, \mathbf{y}$
- 정규직교(orthonormal): 직교이면서 단위벡터인 경우
- 직교는 두 벡터 사이의 각이 90도임을 의미
- 영벡터는 차원이 동일한 모든 벡터들과 직교
- $\mathbb{R}^n$ 에는 최대 n 개의 상호직교(mutually orthogonal)인 영벡터가 아닌 벡터들이 존재

6. Special Kinds of Matrices and Vectors

예제

Example 3.8 (Orthonormal Basis)

The canonical/standard basis for a Euclidean vector space $\mathbb{R}^n$ is an orthonormal basis, where the inner product is the dot product of vectors.

In $\mathbb{R}^2$, the vectors

$$\mathbf{b}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{b}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (3.35)$$

form an orthonormal basis since $\mathbf{b}_1^\top \mathbf{b}_2 = 0$ and $\|\mathbf{b}_1\| = 1 = \|\mathbf{b}_2\|$.

6. Special Kinds of Matrices and Vectors

행렬의 직교성

- 직교행렬(orthogonal matrix): $A^T A = A A^T = I$ 인 행렬 A
- 직교행렬인 경우 $A^{-1} = A^T$ 가 성립하여 역행렬 계산이 쉬움
- 직교행렬을 곱하는 것은 벡터의 길이(length)를 변화하지 않는 변환을 의미

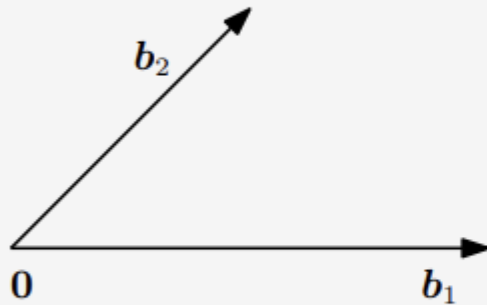
$$\|Ax\|^2 = (Ax)^T (Ax) = x^T A^T A x = x^T I x = x^T x = \|x\|^2$$
- 직교행렬을 곱하는 것은 벡터 사이의 각(angle)을 변화하지 않는 변환을 의미

$$\cos w = \frac{(Ax)^T (Ay)}{\|Ax\| \|Ay\|} = \frac{\|x^T A^T A y\|}{\sqrt{x^T A^T A x y^T A^T A y}} = \frac{(x)^T (y)}{\|x\| \|y\|}$$

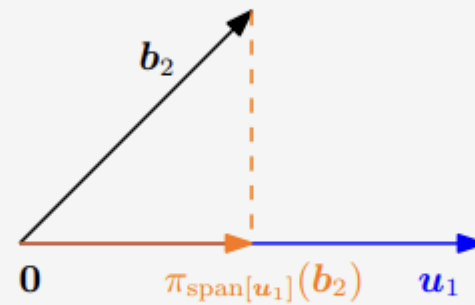
6. Special Kinds of Matrices and Vectors

예제

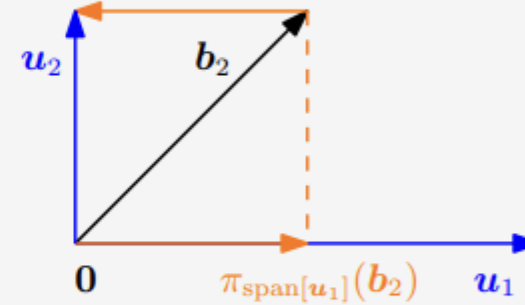
Example 3.12 (Gram-Schmidt Orthogonalization)



(a) Original non-orthogonal basis vectors b_1, b_2 .



(b) First new basis vector $u_1 = b_1$ and projection of b_2 onto the subspace spanned by u_1 .



(c) Orthogonal basis vectors u_1 and $u_2 = b_2 - \pi_{\text{span}[u_1]}(b_2)$.

6. Special Kinds of Matrices and Vectors

예제

Consider a basis $(\mathbf{b}_1, \mathbf{b}_2)$ of $\mathbb{R}^2$, where

$$\mathbf{b}_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \quad (3.69)$$

see also Figure 3.12(a). Using the Gram-Schmidt method, we construct an orthogonal basis $(\mathbf{u}_1, \mathbf{u}_2)$ of $\mathbb{R}^2$ as follows (assuming the dot product as the inner product):

$$\mathbf{u}_1 := \mathbf{b}_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad (3.70)$$

$$\mathbf{u}_2 := \mathbf{b}_2 - \pi_{\text{span}[\mathbf{u}_1]}(\mathbf{b}_2) \stackrel{(3.45)}{=} \mathbf{b}_2 - \frac{\mathbf{u}_1 \mathbf{u}_1^\top}{\|\mathbf{u}_1\|^2} \mathbf{b}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (3.71)$$

7. Eigendecomposition

고윳값

- $n \times n$ 행렬 A 에 대해 $A\mathbf{v} = \lambda\mathbf{v}$ 를 만족하는 스칼라 λ , 영이 아닌 벡터 $\mathbf{v}$ 의 쌍 $(\lambda, \mathbf{v})$ 를 eigenpair라고 함
- λ : 행렬 A 의 고윳값(eigenvalue), 단 λ 가 고윳값이면 $s\lambda$, $s \in \mathbb{R} \setminus \{0\}$ 또한 고윳값
- $\mathbf{v}$: 행렬 A 의 고유벡터(eigenvector)

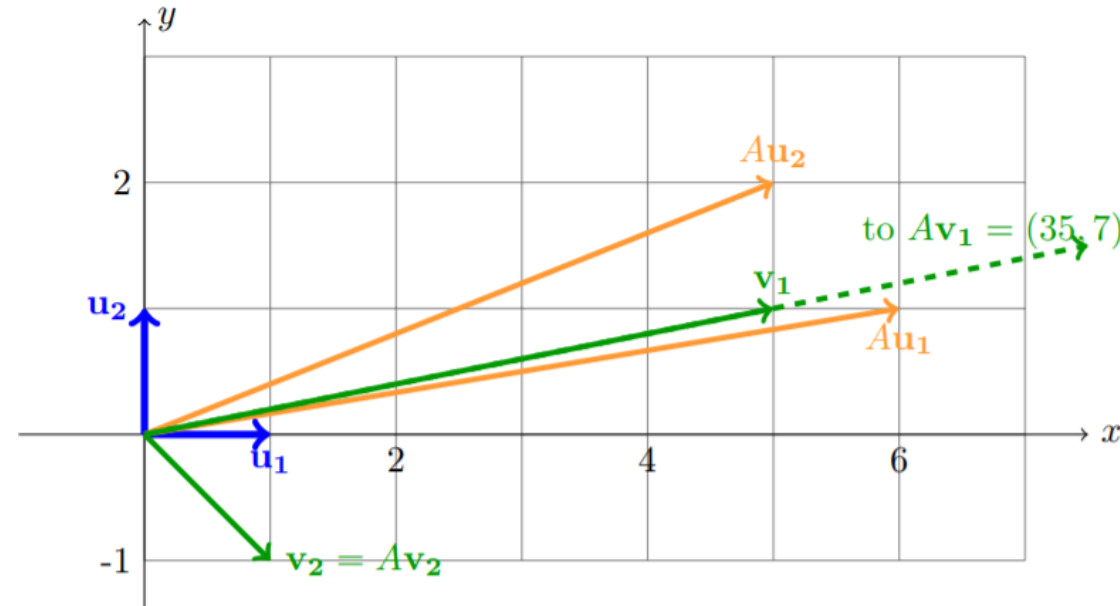
고윳값분해

- $n \times n$ 행렬 A 에 대해 n 개의 선형독립인 고유벡터 $\{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(n)}\}$ 가 존재한다면, 행렬 A 의 고윳값분해(eigendecomposition)는 다음과 같이 표현: $A = V\Lambda V^{-1}$
단, $V = [\mathbf{v}^{(1)} \dots \mathbf{v}^{(n)}]$, $\Lambda = \text{diag}(\boldsymbol{\lambda}) = \text{diag}(\lambda_1, \dots, \lambda_n)$
- $A = V\Lambda V^{-1} \Rightarrow A^k = (V\Lambda V^{-1})^k = V\Lambda^k V^{-1}$
- $\det(A) = \det(V\Lambda V^{-1}) = \det(V) \det(\Lambda) \det(V^{-1}) = \det(\Lambda) = \prod_i \lambda_i$

7. Eigendecomposition

예제

- 행렬은 대부분의 벡터의 크기를 변환(scale)하고 회전(rotate)시키지만, 고유벡터는 크기만 변화시킴
- 고윳값과 고유벡터를 알면 행렬을 이용한 계산을 단순화할 수 있음



Example 4.5 (Computing Eigenvalues, Eigenvectors, and Eigenspaces)

Let us find the eigenvalues and eigenvectors of the 2×2 matrix

$$\mathbf{A} = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}. \quad (4.28)$$

Step 1: Characteristic Polynomial. From our definition of the eigenvector $\mathbf{x} \neq \mathbf{0}$ and eigenvalue λ of $\mathbf{A}$, there will be a vector such that $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$, i.e., $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$. Since $\mathbf{x} \neq \mathbf{0}$, this requires that the kernel (null space) of $\mathbf{A} - \lambda\mathbf{I}$ contains more elements than just $\mathbf{0}$. This means that $\mathbf{A} - \lambda\mathbf{I}$ is not invertible and therefore $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$. Hence, we need to compute the roots of the characteristic polynomial (4.22a) to find the eigenvalues.

Step 2: Eigenvalues. The characteristic polynomial is

$$p_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}) \quad (4.29a)$$

$$= \det \left(\begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = \begin{vmatrix} 4 - \lambda & 2 \\ 1 & 3 - \lambda \end{vmatrix} \quad (4.29b)$$

$$= (4 - \lambda)(3 - \lambda) - 2 \cdot 1. \quad (4.29c)$$

We factorize the characteristic polynomial and obtain

$$p(\lambda) = (4 - \lambda)(3 - \lambda) - 2 \cdot 1 = 10 - 7\lambda + \lambda^2 = (2 - \lambda)(5 - \lambda) \quad (4.30)$$

giving the roots $\lambda_1 = 2$ and $\lambda_2 = 5$.

예제

Step 3: Eigenvectors and Eigenspaces. We find the eigenvectors that correspond to these eigenvalues by looking at vectors $\mathbf{x}$ such that

$$\begin{bmatrix} 4 - \lambda & 2 \\ 1 & 3 - \lambda \end{bmatrix} \mathbf{x} = \mathbf{0}. \quad (4.31)$$

For $\lambda = 5$ we obtain

$$\begin{bmatrix} 4 - 5 & 2 \\ 1 & 3 - 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{0}. \quad (4.32)$$

We solve this homogeneous system and obtain a solution space

$$E_5 = \text{span} \left[\begin{bmatrix} 2 \\ 1 \end{bmatrix} \right]. \quad (4.33)$$

This eigenspace is one-dimensional as it possesses a single basis vector.

Analogously, we find the eigenvector for $\lambda = 2$ by solving the homogeneous system of equations

$$\begin{bmatrix} 4 - 2 & 2 \\ 1 & 3 - 2 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \mathbf{x} = \mathbf{0}. \quad (4.34)$$

This means any vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, where $x_2 = -x_1$, such as $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$, is an eigenvector with eigenvalue 2. The corresponding eigenspace is given as

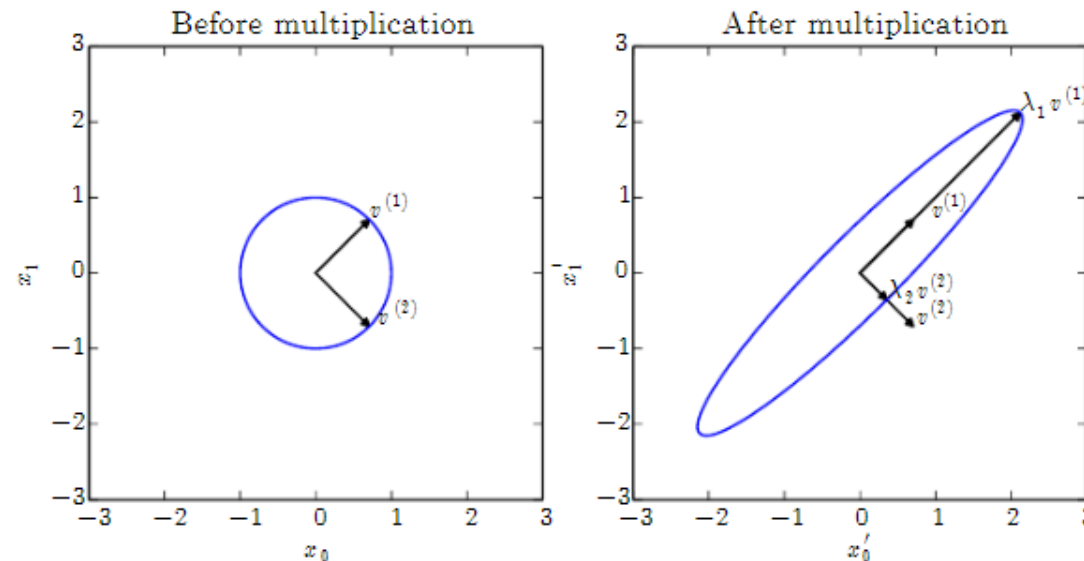
$$E_2 = \text{span} \left[\begin{bmatrix} 1 \\ -1 \end{bmatrix} \right]. \quad (4.35)$$

예제

7. Eigendecomposition

고윳값분해

- 대칭행렬 A 의 고윳값분해는 다음과 같이 표현 가능: $A = Q\Lambda Q^T$
단, Q 는 A 의 직교행렬(orthogonal matrix), 즉 열들이 정규직교(orthonormal)
- A 는 각 고윳값 λ_i 에 대해 연관된 고유벡터 $v^{(i)}$ 방향으로의 공간 크기변환(scaling)



7. Eigendecomposition

이차형식의 극값

- 이차형식(quadratic expression): $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x}$
- $\|\mathbf{x}\|_2 = 1$ 일 때, f 의 최대값은 고윳값의 최대값, 최소값은 고윳값의 최소값과 동일

이차형식의 부호

- 양의 정부호(positive definite): 행렬의 고윳값이 모두 > 0 (≥ 0 인 경우 semi-)
 $\forall \mathbf{x} \neq \mathbf{0} : \mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ 으로 정의 가능
- 음의 정부호(negative definite): 행렬의 고윳값이 모두 < 0 (≤ 0 인 경우 semi-)
 $\forall \mathbf{x} \neq \mathbf{0} : \mathbf{x}^\top \mathbf{A} \mathbf{x} < 0$ 으로 정의 가능
- 두 경우 모두 다음 식이 성립: $\mathbf{x}^\top \mathbf{A} \mathbf{x} = 0 \Rightarrow \mathbf{x} = \mathbf{0}$

7. Eigendecomposition

예제

Example 3.4 (Symmetric, Positive Definite Matrices)

Consider the matrices

$$\mathbf{A}_1 = \begin{bmatrix} 9 & 6 \\ 6 & 5 \end{bmatrix}, \quad \mathbf{A}_2 = \begin{bmatrix} 9 & 6 \\ 6 & 3 \end{bmatrix}. \quad (3.12)$$

$\mathbf{A}_1$ is positive definite because it is symmetric and

$$\mathbf{x}^\top \mathbf{A}_1 \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 9 & 6 \\ 6 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (3.13a)$$

$$= 9x_1^2 + 12x_1x_2 + 5x_2^2 = (3x_1 + 2x_2)^2 + x_2^2 > 0 \quad (3.13b)$$

for all $\mathbf{x} \in V \setminus \{\mathbf{0}\}$. In contrast, $\mathbf{A}_2$ is symmetric but not positive definite because $\mathbf{x}^\top \mathbf{A}_2 \mathbf{x} = 9x_1^2 + 12x_1x_2 + 3x_2^2 = (3x_1 + 2x_2)^2 - x_2^2$ can be less than 0, e.g., for $\mathbf{x} = [2, -3]^\top$.

8. Singular Value Decomposition

특이값분해

- 고윳값분해: $n \times n$ 행렬 A 에 대해, $A = V\Lambda V^{-1}$
- 특이값분해(Singular Value Decomposition, SVD): $m \times n$ 행렬 A 에 대해, $A = U\Sigma V^T$

U : $m \times m$ orthonormal matrix, left-singular vectors of A

V : $n \times n$ orthonormal matrix, right-singular vectors of A

Σ : $m \times n$ diagonal matrix, singular values of A , $\Sigma_{i,j} = \begin{cases} \sigma_i & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$ 이고, $\sigma_i \geq 0$ 만족

- 이 때, U, D, V 는 다음을 만족:

U : AA^T 의 고유벡터들

V : $A^T A$ 의 고유벡터들

Σ : AA^T 의 고윳값들 또는 $A^T A$ 의 고윳값들

$$\begin{matrix} n \\ \boxed{A} \\ m \end{matrix} = \begin{matrix} m \\ \boxed{U} \\ m \end{matrix} \begin{matrix} n \\ \boxed{\Sigma} \\ m \end{matrix} \begin{matrix} n \\ \boxed{V^T} \\ n \end{matrix}$$

8. Singular Value Decomposition

특이값분해

- 특이값분해: $m \times n$ 행렬 A 에 대해, $A = U\Sigma V^T$
- 특이값 방정식: $\forall i = 1, \dots, r : Av_i = \sigma_i u_i$, 즉, $u_i = \frac{1}{\sigma_i} Av_i$
- 특이값: A 의 계수(rank) $r \in [0, \min(m, n)]$ 에 대해, $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r \geq 0$
- $AA^T = (U\Sigma V^T)(U\Sigma V^T)^T = U\Sigma V^T V \Sigma^T U^T = U\Sigma \Sigma^T U^T = U \begin{bmatrix} \sigma_1^2 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_m^2 \end{bmatrix} U^T$
- $A^T A = (U\Sigma V^T)^T (U\Sigma V^T) = V \Sigma^T U^T U \Sigma V^T = V \Sigma^T \Sigma V^T = V \begin{bmatrix} \sigma_1^2 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n^2 \end{bmatrix} V^T$

예제

Example 4.13 (Computing the SVD)

Let us find the singular value decomposition of

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & 0 \end{bmatrix}. \quad (4.81)$$

The SVD requires us to compute the right-singular vectors $\mathbf{v}_j$, the singular values σ_k , and the left-singular vectors $\mathbf{u}_i$.

Step 1: Right-singular vectors as the eigenbasis of $\mathbf{A}^\top \mathbf{A}$.

We start by computing

$$\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 5 & -2 & 1 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}. \quad (4.82)$$

We compute the singular values and right-singular vectors $\mathbf{v}_j$ through the eigenvalue decomposition of $\mathbf{A}^\top \mathbf{A}$, which is given as

$$\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} \frac{5}{\sqrt{30}} & 0 & \frac{-1}{\sqrt{6}} \\ \frac{-2}{\sqrt{30}} & \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{6}} \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{5}{\sqrt{30}} & \frac{-2}{\sqrt{30}} & \frac{1}{\sqrt{30}} \\ 0 & \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{-1}{\sqrt{6}} & \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix} = \mathbf{P} \mathbf{D} \mathbf{P}^\top, \quad (4.83)$$

Step 2: Singular-value matrix.

As the singular values σ_i are the square roots of the eigenvalues of $\mathbf{A}^\top \mathbf{A}$ we obtain them straight from $\mathbf{D}$. Since $\text{rk}(\mathbf{A}) = 2$, there are only two nonzero singular values: $\sigma_1 = \sqrt{6}$ and $\sigma_2 = 1$. The singular value matrix must be the same size as $\mathbf{A}$, and we obtain

$$\Sigma = \begin{bmatrix} \sqrt{6} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}. \quad (4.85)$$

Step 3: Left-singular vectors as the normalized image of the right-singular vectors.

We find the left-singular vectors by computing the image of the right-singular vectors under $\mathbf{A}$ and normalizing them by dividing them by their corresponding singular value. We obtain

$$\mathbf{u}_1 = \frac{1}{\sigma_1} \mathbf{A} \mathbf{v}_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{5}{\sqrt{30}} \\ \frac{-2}{\sqrt{30}} \\ \frac{1}{\sqrt{30}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{bmatrix}, \quad (4.86)$$

$$\mathbf{u}_2 = \frac{1}{\sigma_2} \mathbf{A} \mathbf{v}_2 = \frac{1}{1} \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}, \quad (4.87)$$

$$\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2] = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}. \quad (4.88)$$

예제

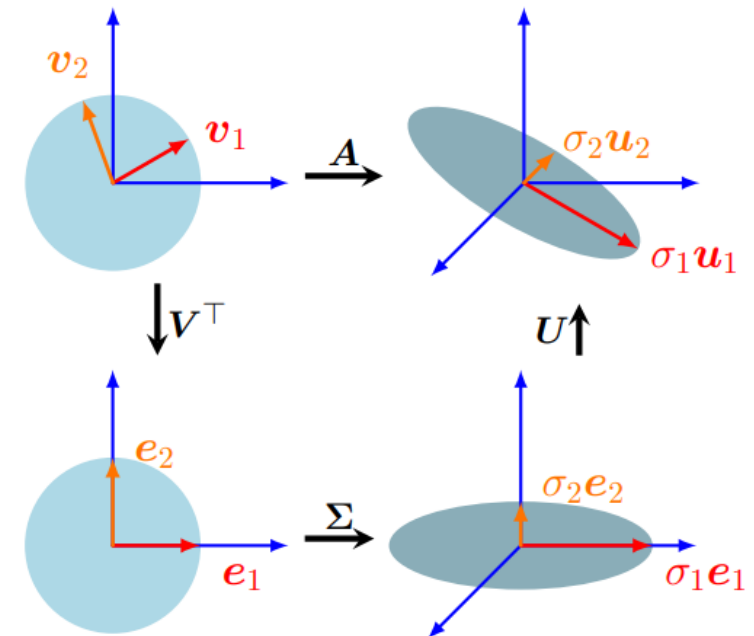
8. Singular Value Decomposition

고윳값분해

- 고윳값분해는 행렬 $A \in \mathbb{R}^{n \times n}$ 에 대해 특성방정식 $\det(A - \lambda I) = 0$ 의 해가 존재하는 경우에만 존재
- 고윳값분해 $A = V\Lambda V^{-1}$ 에서 V 는 직교행렬일 필요가 없음
- Basis 변환 $\rightarrow$ Scaling $\rightarrow$ Basis 변환 수행
- 고윳값은 음수가 될 수 있음

특이값분해

- 특이값분해는 행렬 $A \in \mathbb{R}^{m \times n}$ 에 대해 항상 존재
- 특이값분해 $A = U\Sigma V^T$ 에서 U, V 는 항상 직교행렬
- Basis 변환 $\rightarrow$ Scaling $\rightarrow$ Basis 변환 수행
- 특이값은 항상 음이 아님(non-negative)



8. Singular Value Decomposition

예제

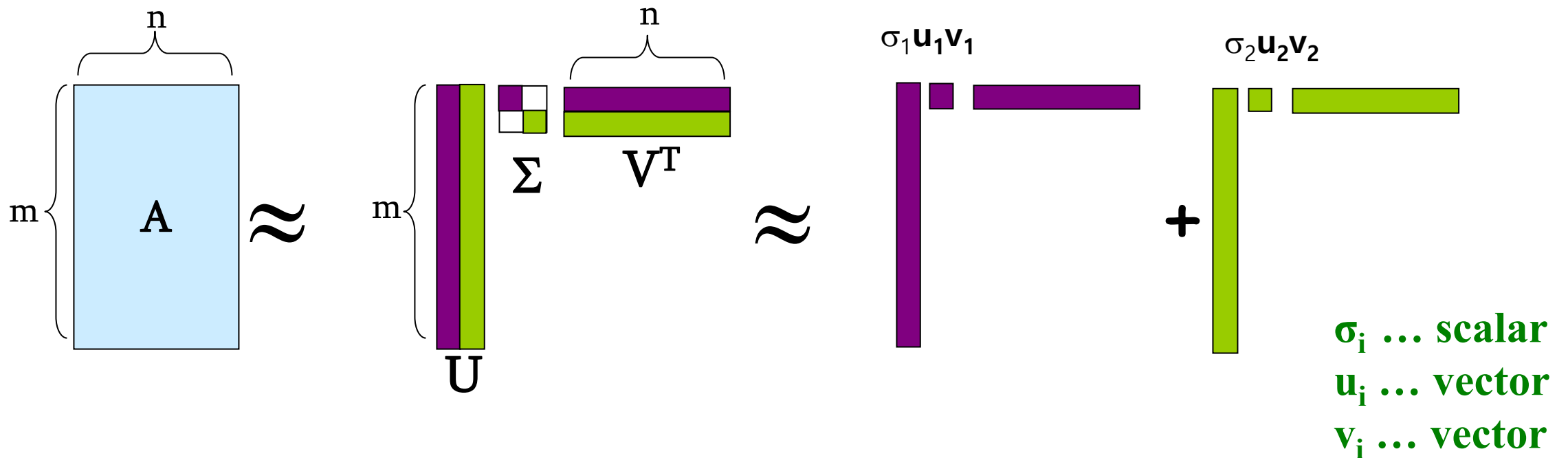
- Finding structure in movie ratings and consumers

$$\begin{array}{c} \text{Star Wars} \\ \text{Blade Runner} \\ \text{Amelie} \\ \text{Delicatessen} \end{array} \begin{array}{c} \text{Ali} \\ \text{Beatrix} \\ \text{Chandra} \end{array} \begin{bmatrix} 5 & 4 & 1 \\ 5 & 5 & 0 \\ 0 & 0 & 5 \\ 1 & 0 & 4 \end{bmatrix} = \begin{bmatrix} -0.6710 & 0.0236 & 0.4647 & -0.5774 \\ -0.7197 & 0.2054 & -0.4759 & 0.4619 \\ -0.0939 & -0.7705 & -0.5268 & -0.3464 \\ -0.1515 & -0.6030 & 0.5293 & -0.5774 \end{bmatrix}$$

$$\begin{bmatrix} 9.6438 & 0 & 0 \\ 0 & 6.3639 & 0 \\ 0 & 0 & 0.7056 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} -0.7367 & -0.6515 & -0.1811 \\ 0.0852 & 0.1762 & -0.9807 \\ 0.6708 & -0.7379 & -0.0743 \end{bmatrix}$$

8. Singular Value Decomposition

$$\mathbf{A} \approx \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T = \sum_i \sigma_i \mathbf{u}_i \circ \mathbf{v}_i^T$$



9. The Moore–Penrose Pseudoinverse

Moore–Penrose 유사역행렬

- $Ax = b$ 라는 문제를 풀기 위해서 A 의 left-inverse인 B 를 양변에 곱할 수 있다면, $x = Bb$
- 행렬 $A \in \mathbb{R}^{m \times n}$ 에 대해 $m > n$ 이면 해가 없을 수 있고, $m < n$ 이면 해가 무수히 많을 수 있음
- 유사역행렬:

$$A^+ = V\Sigma^+U^\top = V \begin{bmatrix} \Sigma_r^{-1} & \mathbf{0}_{r \times (m-r)} \\ \mathbf{0}_{(n-r) \times r} & \mathbf{0}_{(n-r) \times (m-r)} \end{bmatrix} U^\top = (A^\top A)^{-1} A^\top$$

- 유사역행렬은 $\|Ax - b\|_2$ 를 최소로 하는 최적해

10. The Trace Operator

대각합

- 대각합(trace): $\text{Tr}(\mathbf{A}) = \sum_i A_{i,i}$
- Frobenius 노름: $\|\mathbf{A}\|_F = \sqrt{\sum_{i,j} A_{i,j}^2} = \sqrt{\text{Tr}(\mathbf{A}\mathbf{A}^\top)}$
- $\text{Tr}(\alpha\mathbf{A}) = \alpha\text{Tr}(\mathbf{A})$
- $\text{Tr}(\mathbf{A}^\top) = \text{Tr}(\mathbf{A})$
- $\text{Tr}(\mathbf{A} + \mathbf{B}) = \text{Tr}(\mathbf{A}) + \text{Tr}(\mathbf{B})$
- $\text{Tr}(\mathbf{AB}) = \text{Tr}(\mathbf{BA})$
- $\text{Tr}(\prod_{i=1}^n \mathbf{F}^{(i)}) = \text{Tr}(\mathbf{F}^{(n)} \prod_{i=1}^{n-1} \mathbf{F}^{(i)})$, e.g., $\text{Tr}(\mathbf{ABC}) = \text{Tr}(\mathbf{CAB}) = \text{Tr}(\mathbf{BCA})$

11. The Determinant

행렬식

- 행렬식(determinant): $\det(\mathbf{A})$
- $\mathbf{A}$ 가 가역 $\Leftrightarrow \det(\mathbf{A}) \neq 0$
- $\det(\alpha \mathbf{A}) = \alpha^n \det(\mathbf{A})$
- $\det(\mathbf{A}^\top) = \det(\mathbf{A})$
- $\det(\mathbf{A}^{-1}) = 1 / \det(\mathbf{A})$
- $\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B})$
- $\mathbf{T}$ 가 upper-triangular일 때, $\det(\mathbf{T}) = \prod_{i=1}^n T_{i,i}$
- 특성방정식 $p_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I})$
 $= c_0 + c_1 \lambda + \cdots + c_{n-1} \lambda^{n-1} + (-1)^n \lambda^n$
 $\Rightarrow c_0 = \det(\mathbf{A}), c_{n-1} = (-1)^{n-1} \text{Tr}(\mathbf{A})$

Figure 4.2 The area of the parallelogram (shaded region) spanned by the vectors $\mathbf{b}$ and $\mathbf{g}$ is $|\det([\mathbf{b}, \mathbf{g}])|$.

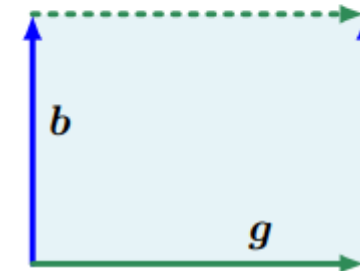
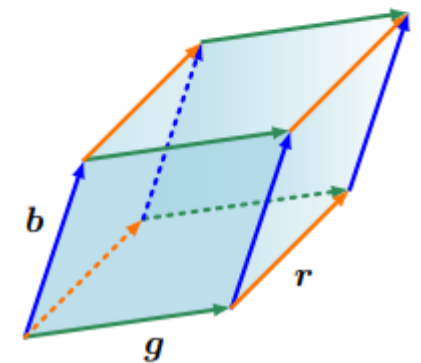


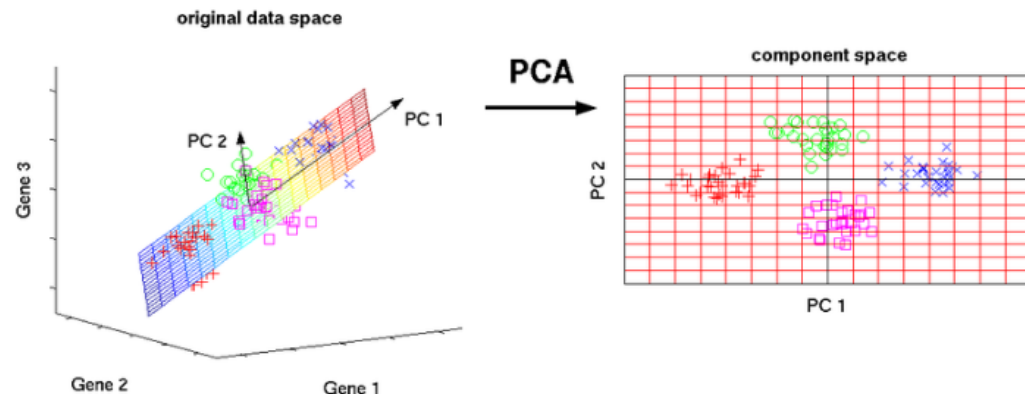
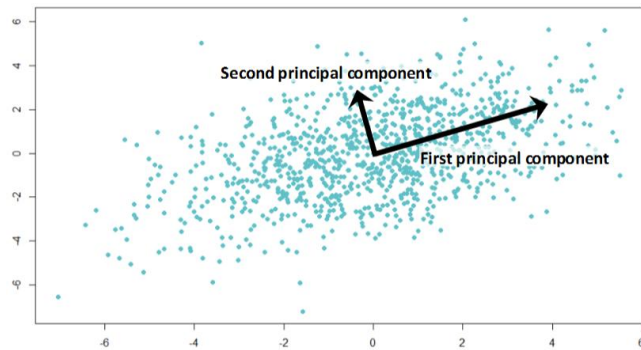
Figure 4.3 The volume of the parallelepiped (shaded volume) spanned by vectors $\mathbf{r}, \mathbf{b}, \mathbf{g}$ is $|\det([\mathbf{r}, \mathbf{b}, \mathbf{g}])|$.



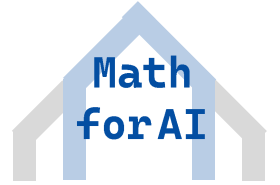
12. Principal Component Analysis

주성분분석

- 차원축소(dimensionality reduction):
고차원 데이터의 불필요한(redundant) 정보를 줄여서 데이터의 차원을 축소하는 방법
- 주성분분석(Principal Component Analysis, PCA):
고차원 데이터를 주성분(principal component)에 투영(projection)하여 차원을 축소하는 방법
- 데이터의 정보를 최대한 유지하면서 차원을 축소하기 위해 분산(variance)이 큰 방향을 순차적 선택



AI를 위한 수학 - Linear Algebra



질문 & 답변



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